

Transformers Provably Learn Algorithmic Solutions for Graph Connectivity

...But Only with the Right Data

Ye, Fu, Jia, Sharan · Preprint Feb 2026 (arXiv 2510.19753v2)

Reading companion · GNN / Graph-Transformer thesis exploration

★ TL;DR

A depth- L Transformer can **provably** solve graph connectivity by computing powers of the adjacency matrix — but **only on graphs whose diameter is $\leq 3^L$** (the model’s **capacity**). Expressivity is not the bottleneck; the **training distribution** is. If too many training graphs exceed capacity ($\text{diam} > 3^L$), gradient descent abandons the matrix-powering algorithm and settles for a brittle **degree-counting heuristic**. The fix is a **data lever**: restrict training to within-capacity graphs and the exact algorithm emerges — and this transfers from the toy **Disentangled Transformer** to standard ones.

♦ The big question

Why do Transformers prefer brittle **heuristics** over **verifiably correct algorithms**, even when the task admits an algorithmic solution and the architecture is expressive enough to represent it?

♦ The one-line answer

Because the **training data** contains too many instances **harder than the model’s exact capacity** (3^L), and those instances reward a global shortcut. Curate the data → recover the algorithm.

1 · Setup — what is actually being learned

► Definition — Task

Input = **self-loop-augmented adjacency matrix** $A \in \{0, 1\}^{n \times n}$ (i.e. $A + I$). Target = **connectivity matrix** R , where $R_{ij} = 1$ iff a path exists between v_i, v_j — equivalently iff $[A^n]_{ij} > 0$. Models map $\mathcal{M} : \{0, 1\}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$; “perfect” means the sign pattern of $\mathcal{M}(A)$ matches R everywhere. Metric: **Exact-Match Accuracy** (whole matrix correct), averaged over Erdős–Rényi $\text{ER}(n, p)$ graphs.

📌 My note

The self-loop trick is the whole game: $[A^k]_{ij}$ counts walks of length **exactly** k but with self-loops a walk can “wait”, so $[A^k]_{ij} > 0$ iff distance $\leq k$. Connectivity therefore = “is some power of A positive here” = **transitive closure via matrix powering** (Warshall/Floyd). This is the algorithmic target the whole paper is chasing.

The architecture: Disentangled Transformer (DT)

A linear read-in XW_{in} , L attention blocks, linear read-out. The twist: each block **appends** its output as new coordinates of the residual stream (instead of summing), so read/write paths stay traceable — making the dynamics analyzable. Attention is $\text{Attn}(h; W) = \frac{1}{n} \text{ReLU}(hWh^\top)h$. The hidden width grows like $d_\ell = 2^{\ell+1}n$. Standard attention-only Transformers can be **re-expressed** as DTs, so results are claimed to transfer (verified empirically, Fig 7 & 10).

📌 My note

The DT is a **mechanistic proxy**, à la Friedman/Nichani. The bilinear $hWh^\top$ form under non-negative weights is what makes the algebra of the two-channel decomposition (§3) go through. Worth keeping in mind how much of the theory leans on the **non-negative weight** assumption — they note training does **not** enforce it, yet predictions still hold.

2 · Capacity — the 3^L reach

◆ Theorem — 4.3 Expressivity (upper reach)

There exists an L -layer DT that is **perfect on every graph with $\text{diam}(G) \leq 3^L$** , by implementing $\sum_{j=0}^{3^L} \alpha_j A^j$ with $\alpha_j > 0$ — the matrix-powering algorithm. (Sketch: set every $W_\ell = I$.)

◆ Theorem — 4.5 Capacity (tight upper bound)

For **any** non-negative weights there is a graph with $\text{diam} = 3^L + 1$ on which the DT fails. So 3^L is exactly the model's **capacity** — it cannot master connectivity beyond path length 3^L , and needs $n = \Omega(3^L)$ nodes to even pose the hard case.

Why **triple** per layer? One attention layer composes a node's features with its neighbors' — reaching 1 hop of new info — but stacked through the appended stream each layer can **cube** the previous reach: $1 \rightarrow 3 \rightarrow 9 \rightarrow \dots \rightarrow 3^L$.

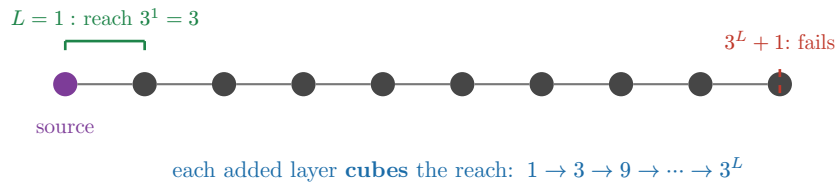


Figure 1: Reach grows as 3^L in graph **diameter**, not in node count n . The decisive depth law is $L \geq \log_3 \text{diam}(G)$ — non-asymptotic and exact, unlike the usual $\Theta(\log n)$.

✍ My note

This **exact, non-asymptotic 3^L** is the paper's lever over prior work (Merrill–Sabharwal give $\mathcal{O}(\exp(L))$ asymptotics). Because the bound is exact, they get a **clean dichotomy** (within- vs beyond-capacity) that drives every later result. The sharpness is the contribution, not the existence of a bound.

3 · The two channels — where the heuristic hides

◆ Theorem — 4.7 Layerwise permutation-equivariant form

Under symmetry of the data ($A \rightarrow R$ commutes with relabeling $PAP^\top \rightarrow PRP^\top$) and non-negative weights, every trained layer weight decomposes as

$$W_\ell = A_\ell \otimes I_n + B_\ell \otimes J_n, \quad A_\ell, B_\ell \in \mathbb{R}^{2 \times 2},$$

where $J_n = \mathbf{1}\mathbf{1}^\top$ is all-ones. Two functionally distinct channels.

⚙ I_n -channel — ALGORITHM

Local. Combines features only between **graph neighbors**. Across layers it composes multi-hop info $\rightarrow$ powers of $A \rightarrow \sum_j \alpha_j A^j$ = exact matrix-powering. **Correct everywhere within capacity.**

⚡ J_n -channel — HEURISTIC

Global, rank-one. $AJ_n = d\mathbf{1}^\top$ broadcasts the **degree vector**. Predicts “connected” from **node degrees** and their products. Correlates with truth on dense ER graphs; **fails adversarially** (two cliques: both high-degree, yet disconnected).

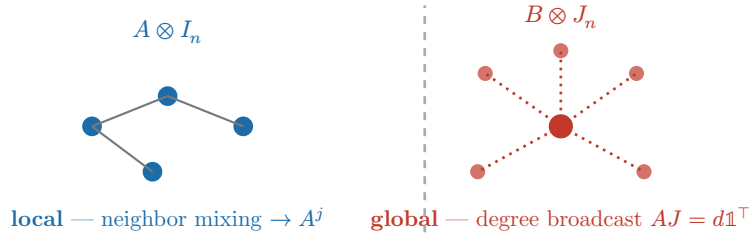


Figure 2: The trained weight is a **superposition** $A_\ell \otimes I_n + B_\ell \otimes J_n$. Learning connectivity = a tug-of-war over how much energy lives in each channel.

4 · Training dynamics — who wins the tug-of-war

◆ Theorem — 4.9 / C.5 / C.9 (dynamics, informal)

Projected gradient descent on the regularized objective converges to KKT points at the standard $\mathcal{O}(\frac{1}{\epsilon})$ rate.

Two phases:

Phase 1 — both channels ramp up on easy within-component pairs (fast, transient, $\sim 2 \cdot 10^2$ of 10^4 steps).

Phase 2 — the data decides. Disconnected/beyond-capacity graphs generate **false positives** in the J -channel and push $B_\ell \rightarrow 0$; connected graphs **reward** it. Net sign of the gradient depends on the **fraction of beyond-capacity graphs**.

◆ Theorem — 4.10 (Learning the Algorithm)

If gradient **penalty** from disconnected graphs outweighs the **reward** from connected ones, the only KKT-compliant value is $B_\ell^* = 0$ — the model converges to the **pure matrix-powering algorithm**.

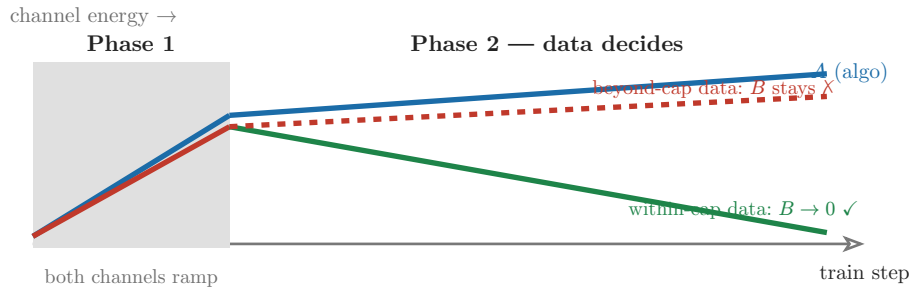


Figure 3: Phase 1 is architecture-driven and identical everywhere. Phase 2 is **data-driven**: the same model forks into the exact algorithm ($B \rightarrow 0$) or a heuristic mixture (B persists) depending on the training distribution.

5 · The data lever — the actionable result

► Definition — 4.6 Within- vs beyond-capacity

A node pair (i, j) is **within capacity** if $[A^{3^L}]_{ij} > 0$ (i.e. $d_{G(i,j)} \leq 3^L$), **beyond capacity** otherwise. A graph is within-capacity if $\text{diam}(G) \leq 3^L$. Define $\rho(\mathcal{G}) = \text{fraction of beyond-capacity node pairs in the distribution}$.

🔧 The lever

Don't train on the raw ER distribution. **Up-weight** (or restrict to) within-capacity graphs $\mathcal{G}_{\leq} = \{G : \text{diam}(G) \leq 3^L\}$. Then the $A \otimes I_n$ channel is promoted to $\sim 100\%$ of the weight energy, the $B \otimes J_n$ heuristic is suppressed, and the model generalizes OOD (e.g. to 2-Chain / 2-Clique graphs it never saw). **Only a small $\rho^* > 0$ is tolerable** — a little beyond-capacity data is fine, a lot is fatal.

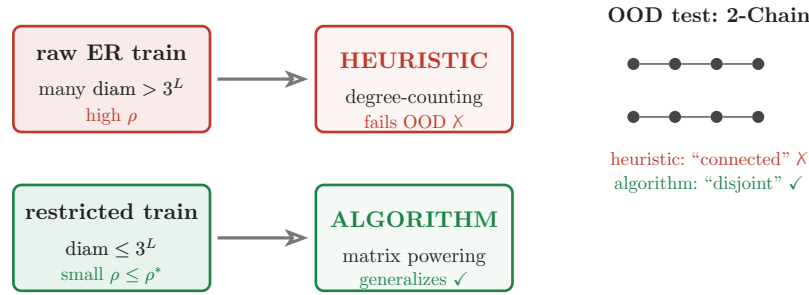


Figure 4: The data lever in one picture. Same architecture, same OOD test (two isolated chains) — outcome flips entirely on the training distribution. The benefit transfers to **standard** Transformers (Fig 7).

6 · Why this matters for a GNN / Graph-Transformer thesis

💡 Thesis hook

Capacity as a curriculum signal. The 3^L bound is exact, so you can **measure** whether each training graph is within/beyond capacity and design a principled **difficulty-aware curriculum** or **rejection-sampling** scheme. Thesis angle: does an analogous exact-capacity law exist for **message-passing GNNs** (reach = L hops, linear not 3^L) and for **sparse / windowed** graph Transformers? The contrast L vs 3^L vs full attention is a clean axis.

💡 Thesis hook

Channel decomposition as a diagnostic. The $A \otimes I + B \otimes J$ split gives a **measurable “heuristic energy”** $\frac{\|B\|}{\|W\|}$. Could become a general **probe** for shortcut-learning in graph models — track it during training across tasks (counting, shortest-path, bipartiteness) and architectures. A lightweight, interpretable thesis contribution.

💡 Thesis hook

Generalize the failure taxonomy. Here the heuristic = degree counting and the adversarial case = two cliques. For other graph tasks (cycle detection, coloring, matching) what is the natural heuristic channel and its adversarial distribution? A thesis could map **task** → **heuristic** → **data-lever** systematically.

💡 Thesis hook

Beyond ER. All theory is on Erdős–Rényi. Real graphs are heavy-tailed / community-structured, where degree heuristics are **stronger**. Does the data lever still recover the algorithm on SBM / power-law / molecular graphs? Strong empirical thesis with a clear hypothesis from this paper.

7 · Open questions & critique

? Open question

Non-negativity gap. Theory assumes $W_\ell \geq 0$; training does not enforce it yet predictions hold (their words). **Why?** A rigorous account of the non-negative case as an attractor would tighten the story — possible thesis sub-problem.

? Open question

Does 3^L survive multi-head / softmax / causal masking? The DT uses single-head ReLU attention, no softmax, no causal mask. How much of the exact capacity is an artifact of these simplifications vs. fundamental?

? Open question

What is ρ^* quantitatively? They show a small tolerable fraction of beyond-capacity data exists, but the threshold depends on graph distribution. Can it be predicted a-priori from (n, p, L) ? That would make the lever a practical recipe rather than a knob to tune.

? Open question

Cost of the lever. Restricting to within-capacity graphs shrinks the usable data and may hurt **length generalization** (Fig 5: $d = 2$ fails to length-generalize). Is there a tension between “learns the algorithm” and “extrapolates to larger graphs”? The sweet spot ($d = \text{Cap}$) seems narrow.

8 · Connections to my reading list

- **Sanford et al. 2024 / 2023** (depth–width tradeoffs, graph tasks): complementary expressivity lower bounds — cross-check the 3^L reach against their communication-complexity arguments.
- **Loukas 2020** (what GNNs cannot learn, depth×width): the GNN analogue of the capacity story — message passing reaches L hops, not 3^L .
- **NeurIPS 2025 depth–width for Transformers on graphs**: directly adjacent; read next, contrast capacity definitions.

Reading status: main body (§1–6, pp. 1–9) read · appendix proofs (B, C) skimmed · next: re-derive Thm 4.7 Kronecker decomposition by hand; pull Fig 6 (ρ sweep) into the ρ^* question. · *Companion generated 2026-06-24.*